

RBE-550 Assignment 1

Turtle Graphics & Obstacle Fields

Victor Sierra Search Pattern and $f(\rho)$
Obstacle Fields

Mohamed Eljahmi



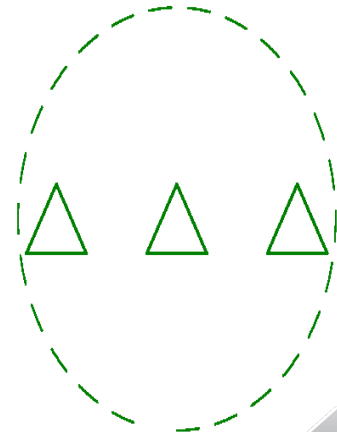
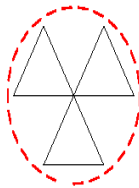
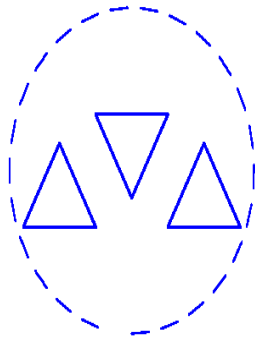
Overview

- Part 1: Draw Victor Sierra search pattern with turtle graphics
- Part 2: Generate obstacle fields with tetrominoes
- Verify outputs against assignment requirements
- Submit code, images, and report



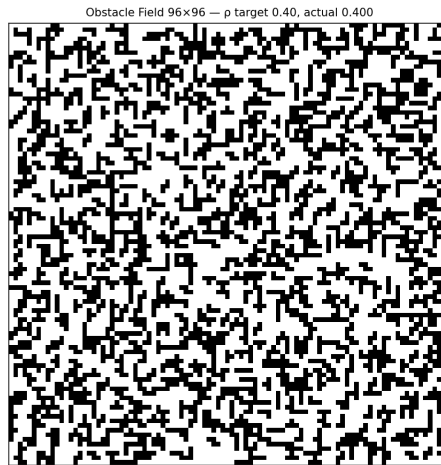
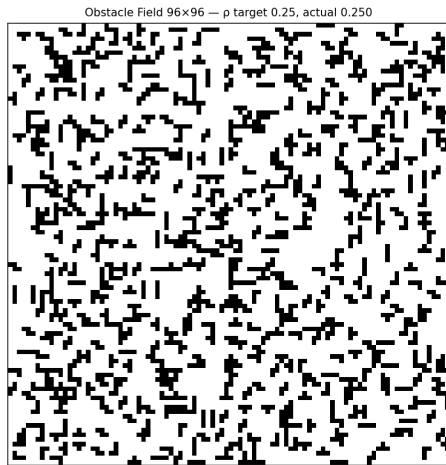
Victor Sierra Search Patterns

- Three layouts implemented with turtle graphics:
 - Zigzag
 - Radial
 - Linear



Obstacle Fields ($f(\rho)$)

- Generated black/white obstacle fields at different coverage:
 - $\rho = 0.25$
 - $\rho = 0.40$
 - $\rho = 0.70$



Environment & Reproducibility

- Python 3.10.12
- Ubuntu 22.04, kernel 6.8.0-78-generic
- Libraries: turtle, numpy, matplotlib
- requirements.txt and environment.txt included



Conclusion

- Implemented Victor Sierra patterns (zigzag, radial, linear)
- Generated obstacle fields at target ρ levels
- Verified actual coverage matches targets
- Submission includes source code, EPS/PNG outputs, README, and report

